

Jiixin Zhang

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Technical Focus: Medical Ultrasound, Ultrasound Image Processing, Medical Imaging AI, Image Reconstruction, Multimodal Imaging, 3D Segmentation, Physics-Informed Simulation, Surgical Guidance

EDUCATION

Johns Hopkins University (JHU) <i>Ph.D Candidate, Electrical and Computer Engineering, GPA 3.96/4.0 PI: Prof. Muyinatu Bell</i>	Baltimore, USA 09/2021 – 12/2026
University of Virginia (UVA) <i>Master of Engineering, Electrical Engineering, GPA 4.0/4.0</i>	Charlottesville, USA 08/2019 – 12/2020
Nanjing University of Science and Technology (NJUST) <i>Bachelor of Engineering, Electrical Engineering, GPA 3.76/4.0</i>	Nanjing, China 09/2015 – 06/2019

PROFESSIONAL EXPERIENCE

Siemens Healthineers <i>Research Intern - Multimodal AI for Medical Imaging</i>	Princeton, USA 06/2026 – present
- Developed a physics-guided, fully differentiable 3D ICE (intracardiac echocardiography) simulator with GPU-based Monte Carlo ray tracing, achieving realistic speckle statistics in 99.4%-100% of myocardial tissue and blood frames - Trained nnU-Net and developed catheter pose-conditioned EchoSAM3D for cardiac segmentation using synthetic 3D ICE datasets generated from PCCT-derived anatomy via differentiable simulation and diffusion model	
United Imaging Healthcare North America, Inc. <i>Ultrasound Engineer Intern</i>	Bellevue, USA 02/2026 – 05/2026
- Developed cardiac ultrasound image processing algorithms using raw RF and IQ channel data, adaptive chamber clutter suppression, coherence- and amplitude-guided enhancement, and temporal filtering across cine sequences - Integrated and validated algorithms on-cart using multi-center <i>in vivo</i> cardiac ultrasound cines across diverse image quality conditions for artifact mitigation and image enhancement	
Ultrasound and Photoacoustic Beamforming <i>Teaching Assistant</i>	Baltimore, USA Spring 2024, Spring 2025
- Delivered lectures on signal processing and image reconstruction for medical ultrasound and photoacoustic imaging - Guided coding labs for acoustic propagation modeling and image formation, and mentored 10+ student projects	
WM Keck Center for Cellular Imaging <i>Research Assistant</i>	Charlottesville, USA 01/2020 – 08/2021
- Developed and integrated machine learning algorithms for cancer cell microscopy data into software FLIMAnalyzer - Built an optical laser scanner with galvanometers and dichroic mirrors into a customized fluorescence microscope	
Sinovation Ventures <i>Research Intern - DeeCamp (AI + Life Science Training Camp and Innovation Challenge)</i>	Beijing, China 06/2018 – 08/2018
- Designed a robotic arm-based smart stacking system with computer vision, winning the Excellent Demo award [Demo] - Applied convolutional neural networks for real-time handwritten digit recognition paired with an RGB-D camera	

RESEARCH PROJECTS

Medical Ultrasound and Photoacoustic-Guided Surgical Interventions for Healthcare <i>Graduate Research, JHU</i>	12/2021 – present Baltimore, USA
Aim: Develop multimodal medical ultrasound-photoacoustic imaging with AR- and deep learning-guided reconstruction	
- Designed a novel imaging pipeline integrating ultrasound and photoacoustic RF data, augmented reality-driven 3D point clouds, and PCA to enable flexible transducer shape fitting for accurate surgical tool visualization - Developed a deep learning-based transducer shape optimization model using a cross-correlation functions and a coherence-driven loss, improving robustness across computational modeling, phantom tests, and clinical breast data - Derived physics-informed models for photoacoustic imaging with laser optical excitation inside soft biological tissues, integrating coherence-based image formation with experimental phantom and <i>ex vivo</i> tissue validation - Achieved 99% optical target depth accuracy in photoacoustic images and 98% shape estimation accuracy with high image contrast of 24 dB. Results were published in IUS 22, SPIE PW 23, BOE 23, SPIE PW 24, IUS 24, JHU News	
Sound Velocity Estimation in Human Breast Ultrasound Imaging <i>Graduate Research, JHU</i>	02/2024 – 01/2026 Baltimore, USA
Aim: Apply spatial coherence features and physics-guided modeling to estimate sound speed for plane wave ultrasound	

- Designed a novel framework that estimates local speed of sound by optimizing short-lag spatial coherence (SLSC) after angular compounding with multiple plane-wave transmissions
- Processed large-scale ultrasound datasets with raw RF channel data and metadata collected from simulations, tissue-mimicking phantoms, and *in vivo* human patient breast tissues
- Achieved 98%-100% estimation accuracy, reduced image quality deviation from 3.4% to 0.6% for *in vivo* human soft tissues than conventional method. Results were published in **IUS 24, Ultrasonics 25, IEEE T-USON 25**

Overfit Detection for Deep Learning-Based Ultrasound Image Reconstruction

09/2021 – 12/2024

Graduate Research, JHU

Baltimore, USA

Aim: Detect overfitting and failure modes in DNNs trained for biomedical ultrasound image reconstruction

- Designed synthetic ultrasound RF raw channel signals to detect overfitting and analyze generalization in DNNs
- Quantified degradation in speckle and lesion structural patterns under progressive data removal
- Demonstrated overfitting behavior with qualitative image comparison between network-produced images and ground truths and metrics of $NCC \leq 0.8$ and $SSIM \leq 0.1$. Results were published in **IUS 22, UITC 25, Ultrasonics 25, [Git](#)**

Machine Learning for Cancer Cell Metabolic Measure with Fluorescence Microscopy Imaging

01/2020 – 11/2024

Graduate Research, UVA

Charlottesville, USA

Aim: Detect early drug responses of HeLa and prostate cancer cells using autoencoder features from microscopy data

- Built a multimodal microscopy learning framework combining fluorescence lifetime imaging microscopy (FLIM) images, fluorescence decay curves, and autoencoders to extract compressed features for cancer cells
- Performed statistical analysis on latent-space features to quantify temporal apoptosis changes after drug treatment
- Achieved improved sensitivity to predict HeLa and cancer cell metabolic activity compared to PCA-based and SOTA method (reduced from 60 mins to 30 mins). Results were published in **SPIE PW 21, SPIE PW 22, J. Biophotonics 25**

CT-to-Ultrasound Image Synthesis of Vertebral Bone Structures

02/2023 – 05/2023

Graduate Project, JHU

Baltimore, USA

- Trained CycleGAN with U-Net generators for cross-modal synthesis of vertebral bone structures from CT to ultrasound
- Optimized cycle-consistency loss and dataset preprocessing to improve image fidelity, model generalization, and AI-driven computational imaging accuracy

SKILLS

Programming & Software: MATLAB, Python, C/C++, Git, Slurm

ML & Imaging Tools: PyTorch, NumPy, OpenCV, Pandas, Mitsuba, Dr.Jit, k-Wave, Field-II

SELECTED PUBLICATIONS

1. **Zhang J**, Zhang Y, Comaniciu D, Sharma P, Physics-Based Differentiable Full-Path Ray Tracing for Patient-Specific 3D ICE Simulation, 2026
2. **Zhang J**, Bell MAL, Augmented Reality-Assisted Flexible Transducer Shape Estimation for Medical Photoacoustic Imaging, 2026, under review
3. **Zhang J**, Zhu Y, Bell MAL, Spatial Coherence-Based Sound Speed Estimation in Plane Wave Ultrasound Imaging, *IEEE Transactions on Ultrasonics*, 2025
4. **Zhang J**, Bell MAL, Overfit Detection Method for Deep Neural Networks Trained to Beamform Ultrasound Images, *Ultrasonics*, 2025
5. **Zhang J**, Wallrabe H, Siller K, Mbogo B, Cassidy T, Alam SR, Periasamy A, Measuring Metabolic Changes in Cancer Cells Using Two-Photon Fluorescence Lifetime Imaging Microscopy and Machine-Learning Analysis, *Journal of Biophotonics*, 2024
6. **Zhang J**, Zhu Y, Bell MAL, Coherence-Based Optimization Using Cumulative Spatial Lags to Estimate Sound Speed in Plane Wave Images of Coherent and Incoherent Targets, *IEEE UFFC-JS*, 2024 (Oral)
7. **Zhang J**, Wiacek A, Feng Z, Ding K, Bell MAL, Flexible Array Transducer for Photoacoustic-Guided Interventions: Phantom and *Ex Vivo* Demonstrations, *Biomedical Optics Express*, 2023
8. **Zhang J**, Wiacek A, González E, Feng Z, Ding K, Bell MAL, A Flexible Array Transducer for Photoacoustic-Guided Surgery, *IEEE IUS*, 2022 (Oral) (Travel Award)

ACADEMIC ACTIVITIES

Invited Talk: International Symposium on Ultrasonic Imaging and Tissue Characterization (UITC), “Deep Neural Networks for Medical Ultrasound Imaging”, Silver Spring, MD, April 29, 2025

Invited Talk: Annual FLIM, FRET and FLIM (Metabolic Imaging) Microscopy Workshop, “Cancer Cell Metabolic Prediction using Machine Learning in FLIM”, Charlottesville, VA, March 6, 2023

Reviewer: Journal of Physics: Photonics, Medical Physics, Physics in Medicine & Biology, Journal of Medical Imaging